Thermal Response Functions for Chikungunya Transmission Risk

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Introduction

Before building the reproduction number model, I first examine the individual temperature-dependent functions used in the model. The purpose is to make sure each function behaves as expected over the temperature range. After checking and approving these functions separately, I will use them in the full $R_0(T)$ model. We use the Mordecai et al. model as the baseline. In this model, the basic reproduction number depends on temperature:

$$R_0(T) = \sqrt{\frac{a(T)^2 b(T) c(T) e^{-(\frac{\mu(T)}{PDR(T)})} EFD(T) p_{EA}(T) MDR(T)}{N r \mu(T)^3}}$$

The temperature-dependent functions are:

$$a(T) = \text{biting rate}$$

$$b(T) = \text{vector-to-human transmission probability}$$

$$c(T) = \text{human-to-vector transmission probability}$$

$$\mu(T) = \text{adult mosquito mortality rate}$$

$$PDR(T) = \text{parasite development rate}$$

$$EFD(T) = \text{eggs per female per day}$$

$$pEA(T) = \text{egg-to-adult survival probability}$$

$$MDR(T) = \text{mosquito development rate}$$

Biting rate

a is the per-mosquito biting rate.

- The Mordecai biting-rate functions were evaluated only where they are positive: $13.35 < T < 40.08$ for *Ae. aegypti* and $10.25 < T < 38.32$ for *Ae. albopictus*.
- Ae. albopictus biting rate is $\sqrt{(38.32 - T)}$ not $\sqrt{(38.31 - T)}$

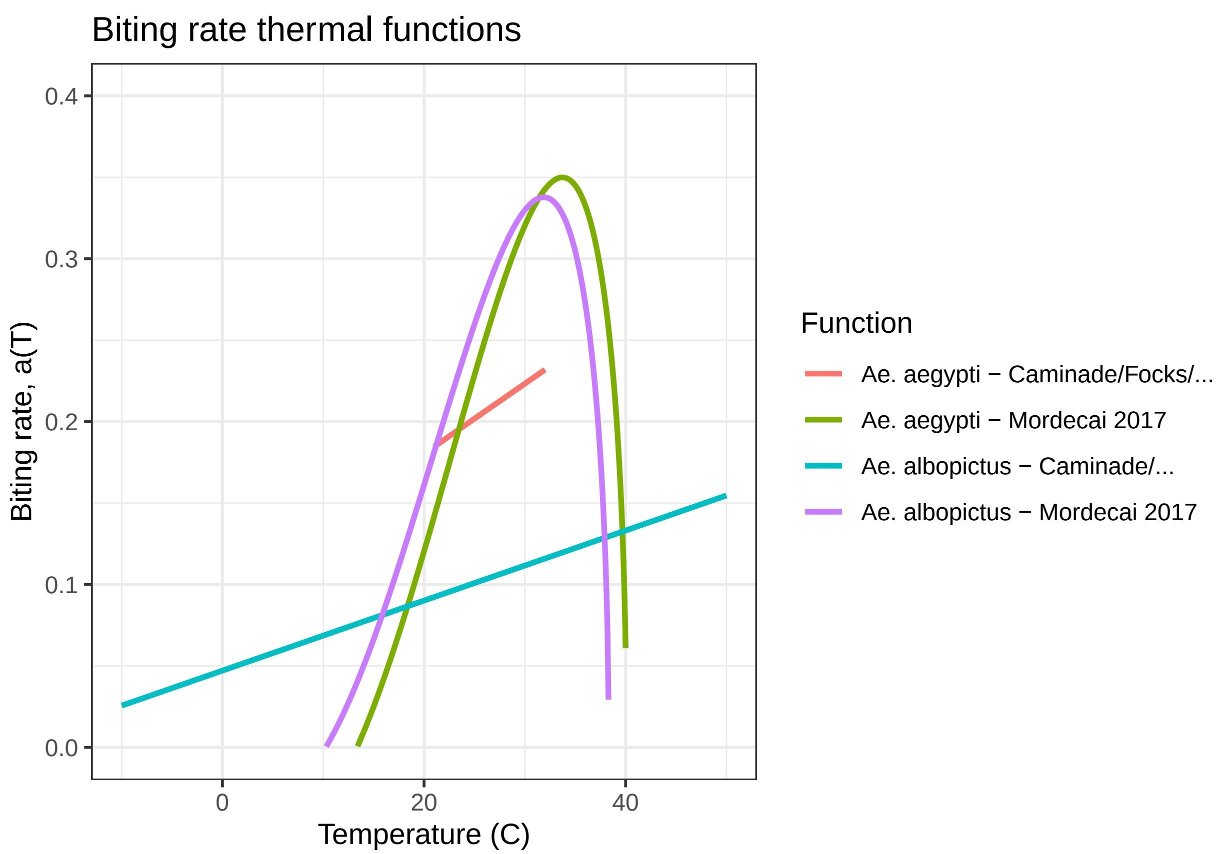

Figure 1: Biting rate thermal functions

Transmission Probability (Vector -> Human)

b is the proportion of infectious bites that infect susceptible humans.

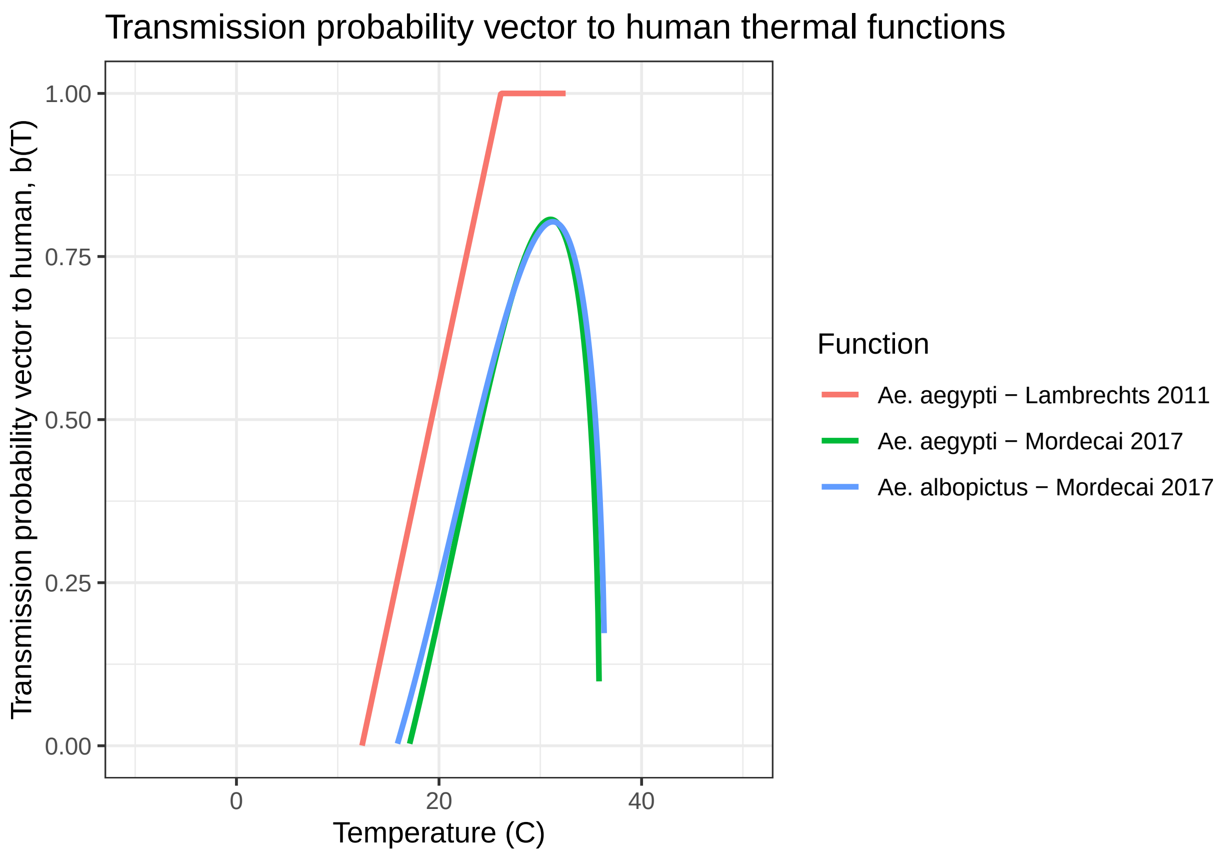

Figure 2: Transmission probability vector to human thermal functions

Transmission Probability (Human -> Vector)

c is the proportion of bites on infected humans that infect previously uninfected mosquitoes. (i.e., b*c = vector competence).

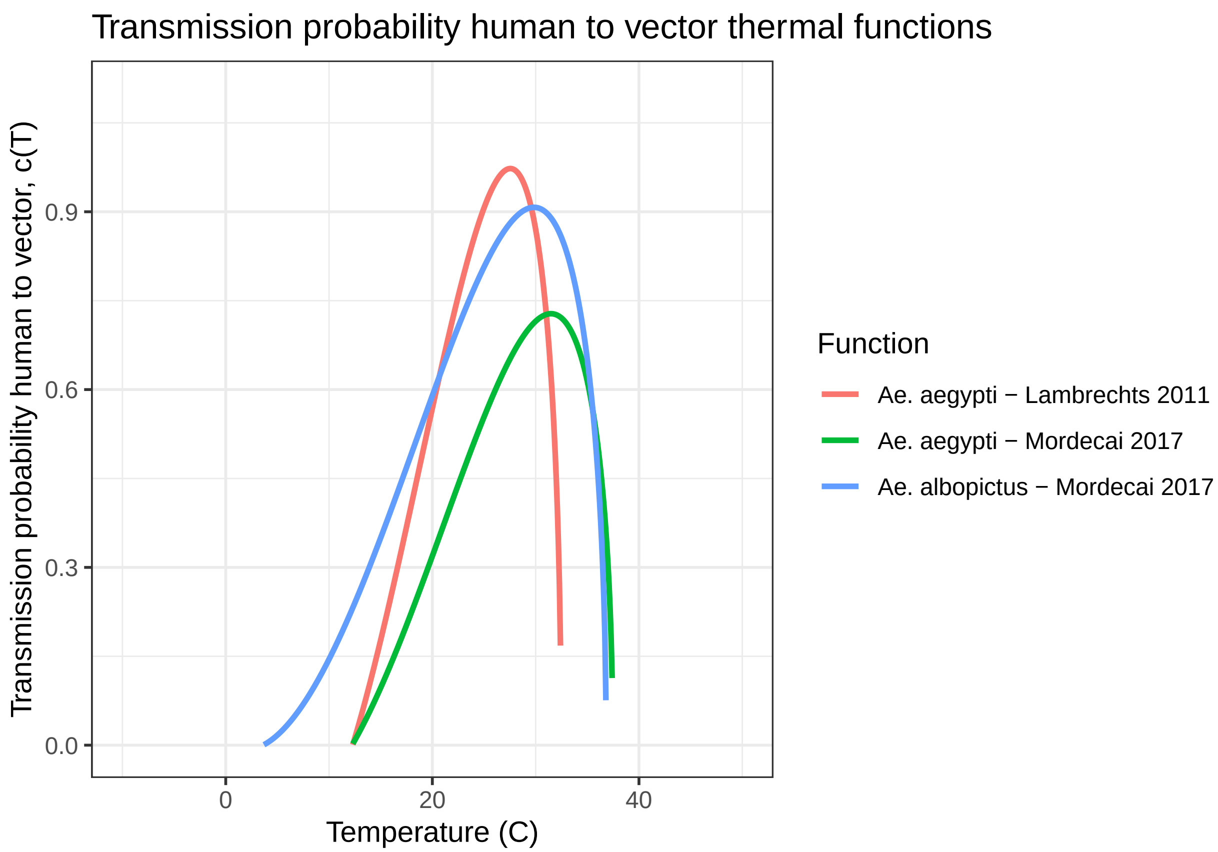

Figure 3: Transmission probability human to vector thermal functions

Adult Mosquito Mortality Rate

μ is the adult mosquito mortality rate.

For Mordecai et al. 2017, the table gives adult mosquito lifespan, $lf(T)$, not mortality rate directly. Therefore, mortality is calculated as: $\mu(T) = \frac{1}{lf(T)}$

- For Ng et al. 2017, the original source was checked. The T^5 coefficient is $-2.72000e^{-8}$, not $-2.72e^{-2}$.
- Mordecai Ae. aegypti functions is $(T - 37.73)$ not $(37.73 - T)$
- Mordecai albopictus functions is $(T - 31.51)$ not $(31.51 - T)$

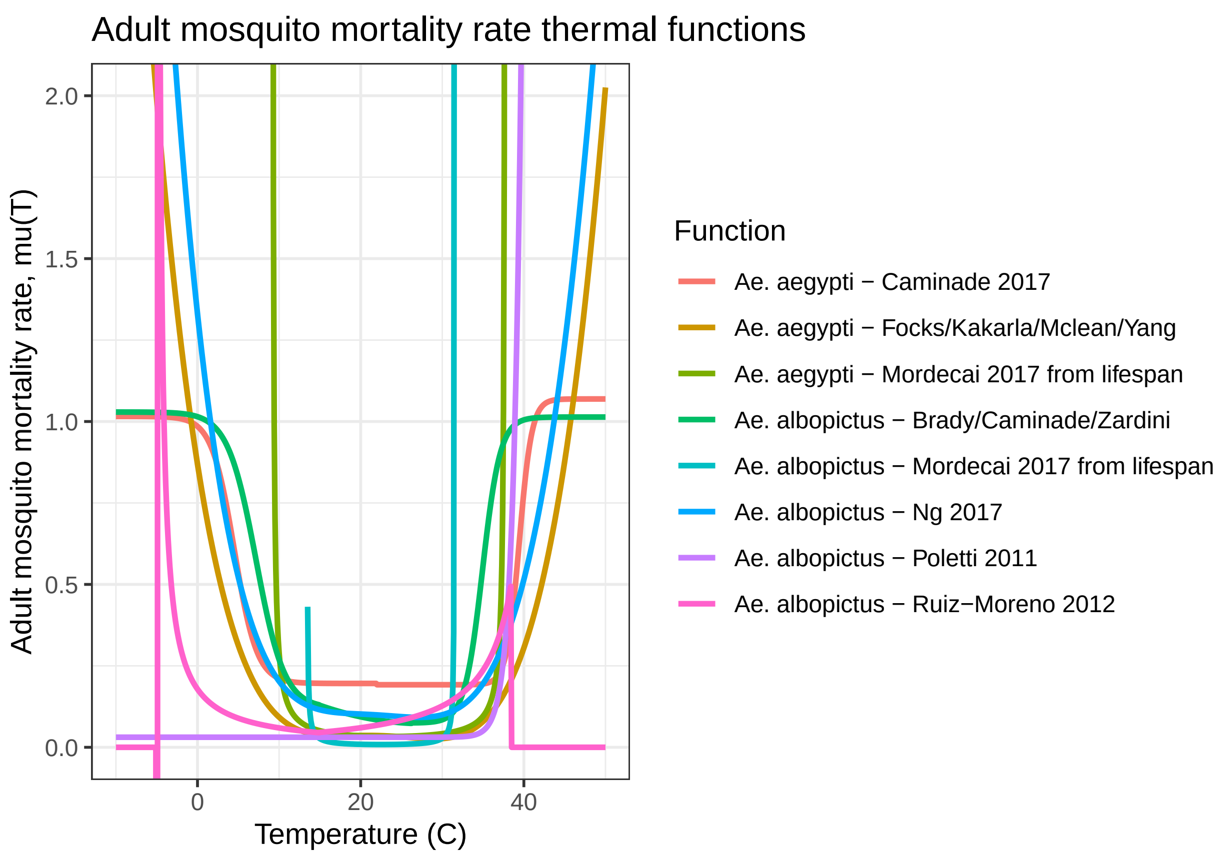

Figure 4: Adult mosquito mortality rate thermal functions

Parasite Development Rate

PDR is the parasite development rate (i.e., the inverse of the extrinsic incubation period, the time required between a mosquito biting an infected host and becoming infectious)

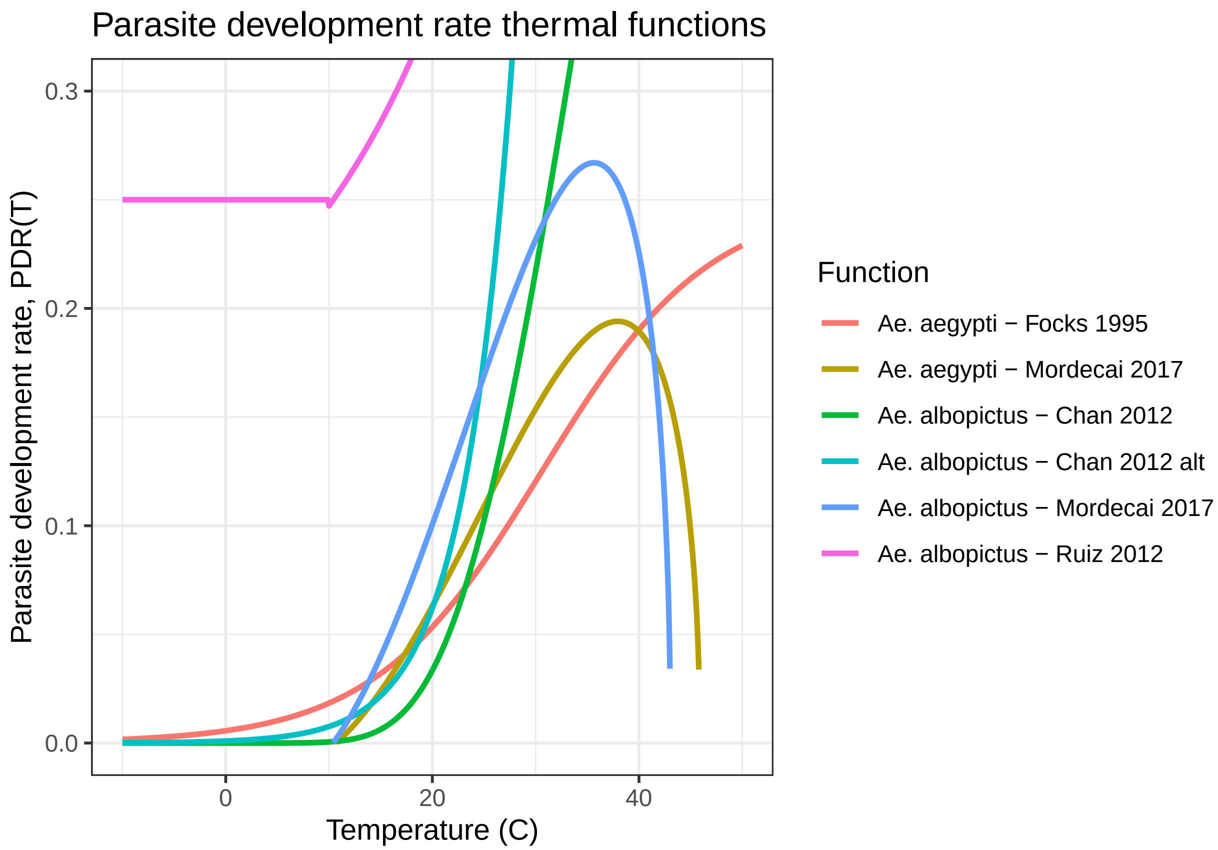

Figure 5: Parasite development rate thermal functions

Eggs per Female per Day (Fecundity)

EFD (Ae. aegypti) is the number of eggs produced per female mosquito per day and TFD (Ae. albopictus) is number of eggs laid per female per gonotrophic cycle (number/female)

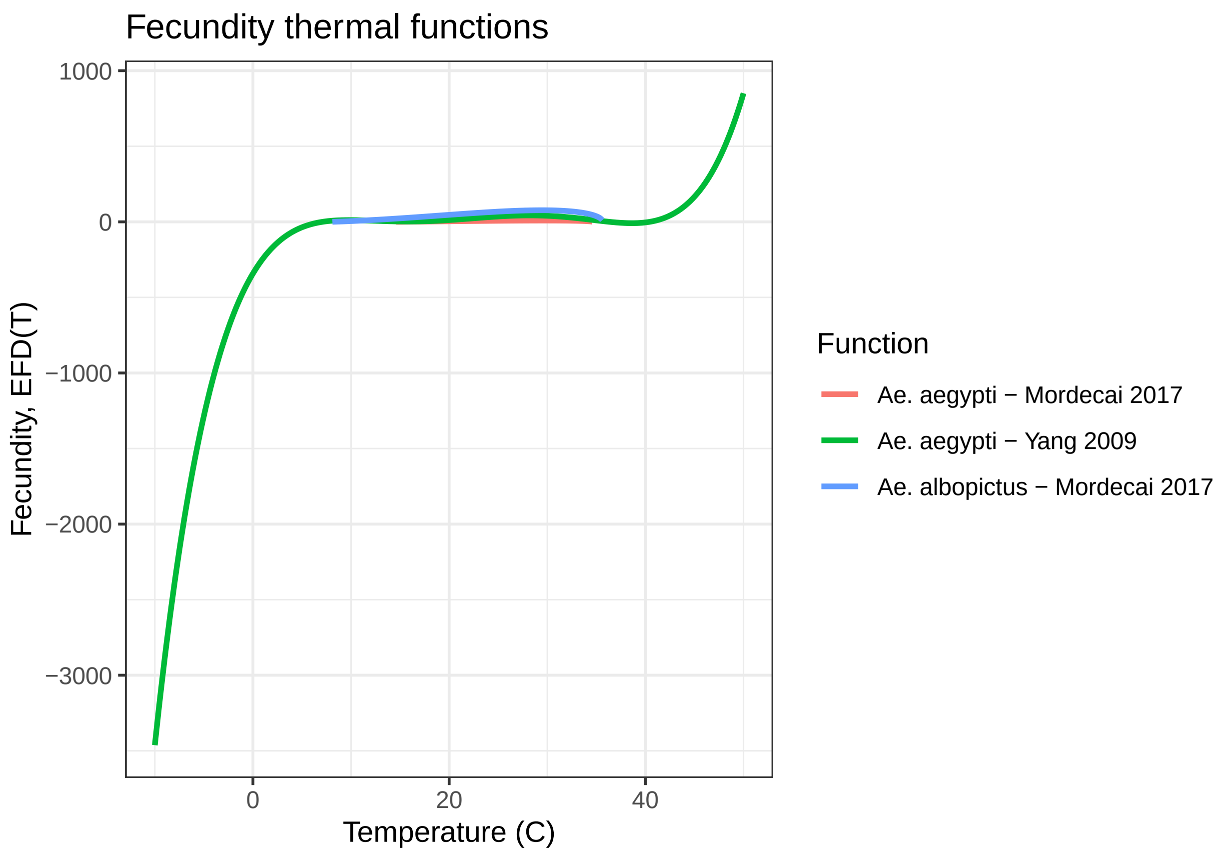

Figure 6: Fecundity thermal functions

Egg-Adult Survival Probability

p_{EA} is the mosquito egg-to-adult survival probability

- The *Ae. albopictus* Mordecai function is $(T - 39.33)$ not $(39.33 - T)$.

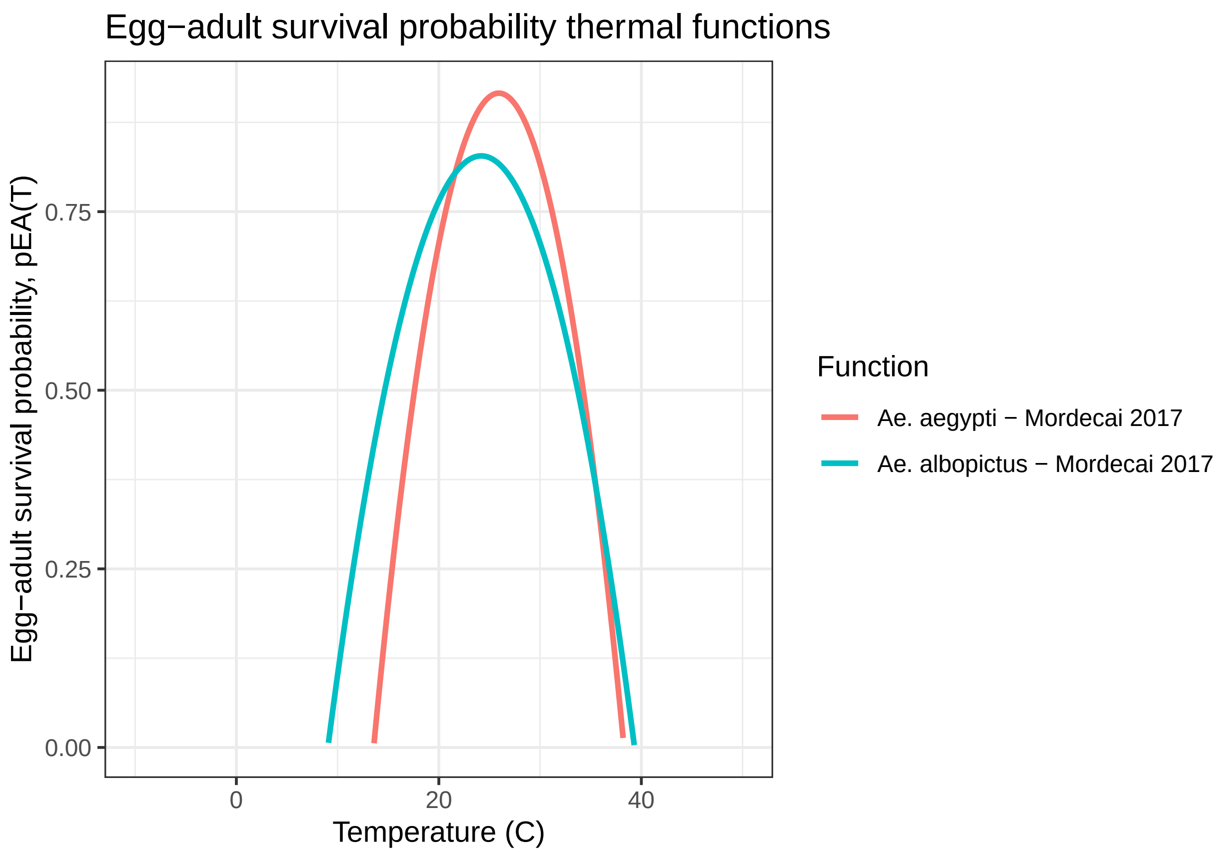

Figure 7: Egg-adult survival probability thermal functions

Mosquito Immature Development Rate (Aquatic Maturation)

MDR is the mosquito immature development rate (i.e., the inverse of the egg-to-adult development time)

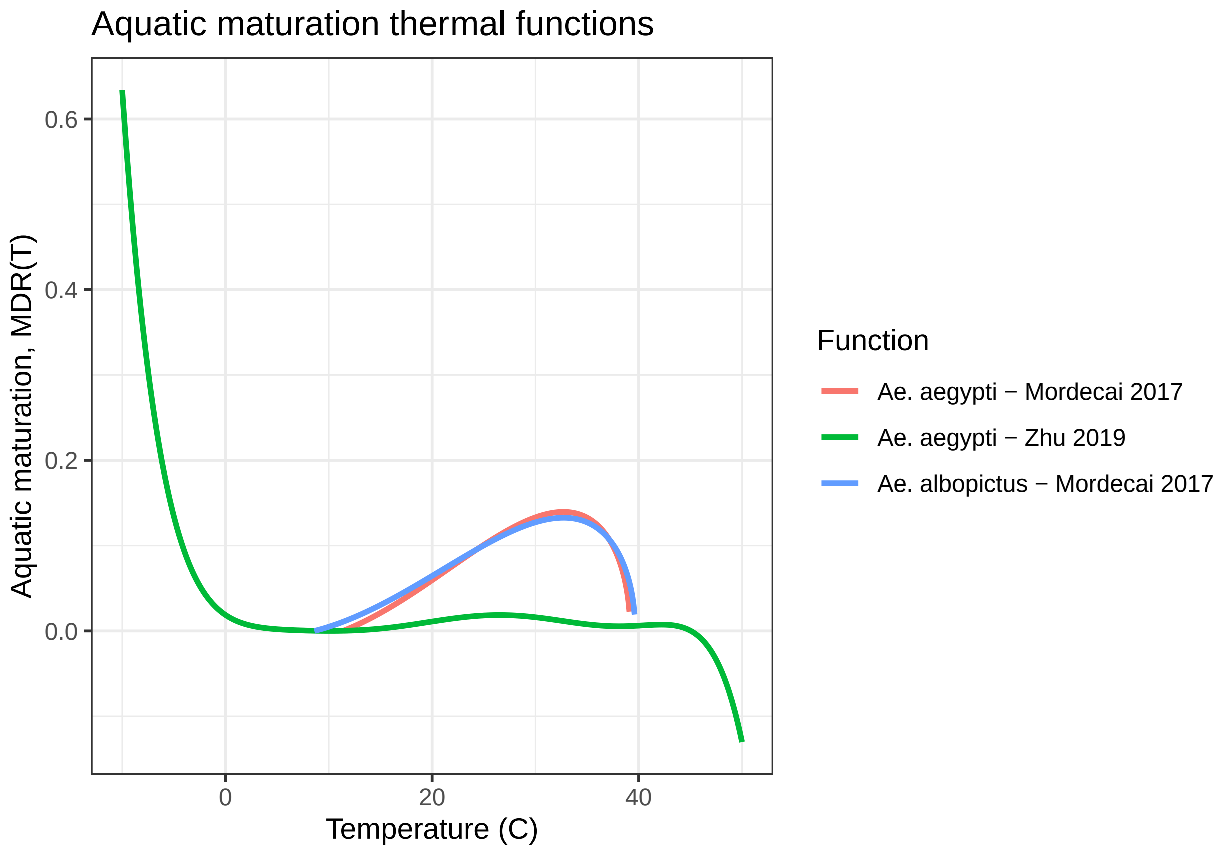

Figure 8: Aquatic maturation thermal functions